

Mobile Application Development with Flutter

Lecture 12: Object Oriented Programming in Dart

OOP in Dart

- **Object-oriented programming (OOP)** is a programming method that uses objects and their interactions to design and program applications
- It is one of the most popular programming paradigms
- In **OOP**, an object can be anything, such as a person, a bank account, a car, or a house
- Each object has its attributes (or properties) and behavior (or methods)
- For example, a person object may have the attributes **name**, **age**, and **height**, and behavior **walk** and **talk**

Class in Dart

- A class defines the properties and methods that an object will have
- An object is an instance of a class
- For example, a class called **Dog** might have properties like **breed**, **color** and methods like **bark**, **run**
- Declaring Class in Dart

```
class ClassName {  
  // properties or fields  
  // methods or functions  
}
```

Creating Classes

- Creating first class
- Constructor
- this keyword
- Constructor short form
- Constructor with optional and named parameters
- Constructor with default values
- Named constructor

Creating Classes

- Encapsulation in Dart
- Properties in Dart
- Set property with condition

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